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Post-Middle Miocene rotations recorded in the Bourg d'Oisans area (Western Alps, France) by paleomagnetism

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Abstract

In order to test the occurrence of post metamorphic rotations or tiltings, a paleomagnetic study was undertaken in 44 sites from the inverted half graben of Bourg d'Oisans ("Zone Dauphinoise", Western Alps). The magnetization carried by pyrrhotite was acquired after metamorphic culmination, around the Oligo–Miocene limit. No obvious rotations around a vertical axis, are recorded. Instead, the dispersion of paleomagnetic data is interpreted as due to a variable tilting around a horizontal N20E axis, consistent with structural observations. This gentle folding in the sedimentary cover is probably associated with thrusting of the Belledonne external crystalline massif over the foreland basement. The age of the tiltings reported in this paper is estimated as being younger than 9–13 Ma.

1. Introduction

Characteristic remanent magnetization (ChRM) of late metamorphic age, can be used as a tectonic marker of the late history of a collision belt. In the Western Alps, the discovery of such a paleomagnetic signal carried by pyrrhotite (Rochette and Lamarche, 1986; Lamarche et al., 1988) allows us to test the existence of rotations around vertical or horizontal axes which occured after the cooling of these rocks. Pyrrhotite derives from the transformation of sedimentary magnetite during alpine metamorphism. Conditions for such a transformation (Rochette and Lamarche, 1986) are particularly favourable in Liassic black schists and limestones of the "Zone Dauphinoise" (Western Alps) (Fig. 1).

Paleomagnetic evidence of post-metamorphic rotations may possibly be linked with: (1) a hypothetical rotation of the external crystalline massifs of 15 to 20 degrees associated with the differential shortening in the French "Chaînes Subalpines" (Gratier et al., 1989; Vialon et al., 1989) or (2) the occurrence of large scale folding comparable to what was found in the area of Maurienne (Ménard and Rochette, 1992 and Lamarche et al., 1988). The area investigated in this study, the Bourg d'Oisans half graben, was chosen to complete in southward the previous studies and to precise on a smaller scale the work of Lamarche et al. (1988).

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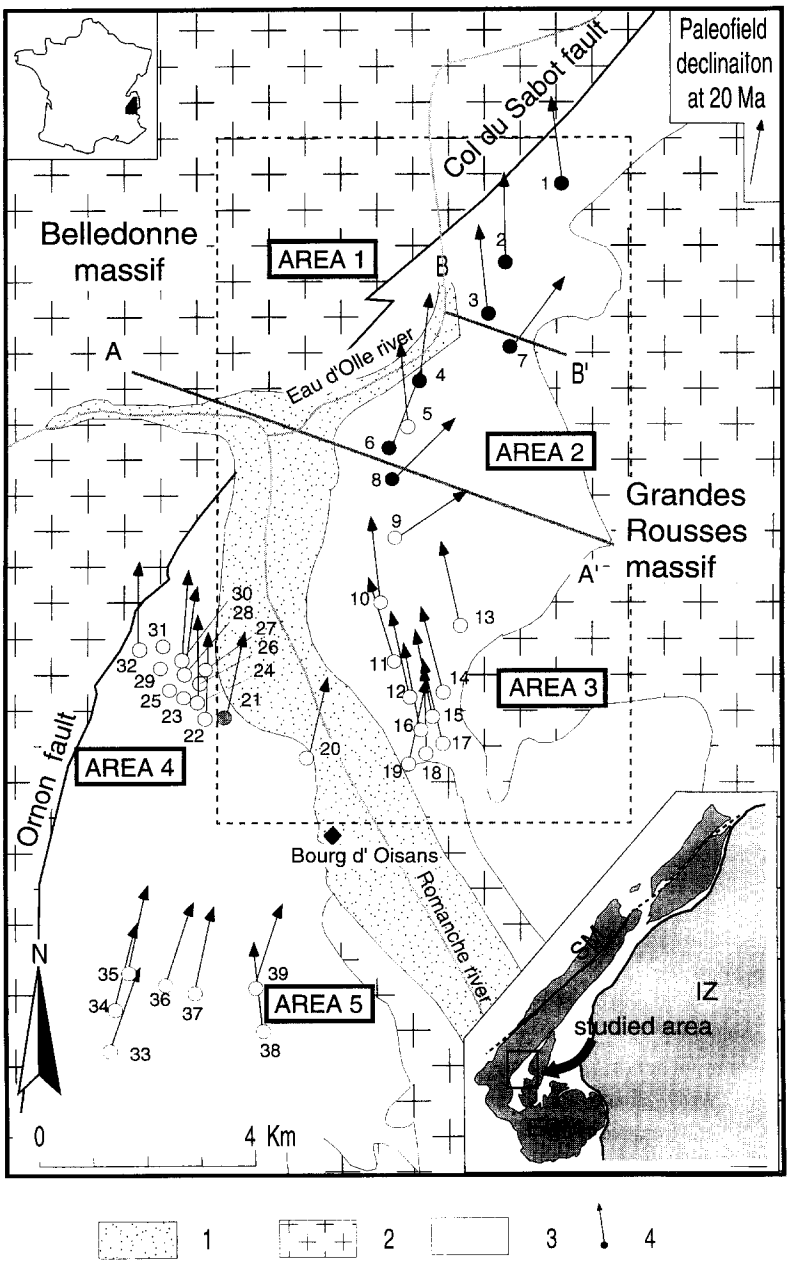


Fig. 1. Local geological sketch map with the site mean magnetization declinaison. *1* Bourg d'Oisans quaternary valley; *2* Crystalline basement; *3* Liassic sedimentary cover; *4* site mean magnetization declinaison (all represented in normal polarity), black: normal polarity; white: reverse polarity; grey: normal and reverse polarity in the same site. Rectangle in grey dashed line is the border of the map of Fig. 5: *AA'* and *BB'*: location of cross sections presented in Fig. 6 and 7 respectively; *ECM*: external crystalline massifs, *IZ*: alpine internal zones, *SM*: "synclinal médian" fault.

2. Geological setting

The studied area covers the inverted half graben of Bourg d'Oisans, between the Belledonne and the Grandes Rousses crystalline massifs. The Bourg d'Oisans half graben, mainly constituted of Liassic marly limestones, is bound to the west by the Triassic–Liassic Ornon–Col du Sabot normal fault (Fig. 1). This area has undergone intense deformation with a slaty cleavage associated with N–S oriented folds (Gratier et al., 1973; Vialon, 1974; Gratier and Vialon, 1980; Gillcrist et al., 1987). This folding is younger than the "Flysch des Aiguilles d'Arves" deposit wich is Priabonian in age (Barbier, 1948). It is generaly known that the age of the regional deformation is Oligocene (Gratier et al., 1973). The deformation is probably contemporaneous than the metamorphism. In the studied area, this one had been dated by Nziengui (1993) at around 26.5 Ma by the K–Ar method on < 2 μm clayey fraction of the metasediments. This suggests that the metamorphism, and so the deformation, occurs during the upper Oligocene.

The synsedimentary normal faults have been reactivated in a complex manner during the alpine tectonics. Dextral strike-slip motion along the Ornon–Col du Sabot fault or along the "synclinal médian" fault are probable but not yet clearly demonstrated on the field (Fig. 1).

The alpine metamorphism in the studied area is recorded by several methods and authors (e.g. Poty, 1969; Gratier et al., 1973; Le Fort, 1973; Poty et al., 1974; Bernard, 1978; Jenatton, 1981; Aprahamian, 1988). The study of Julien and Goffé (1993) in the eastern Dauphinoise zone, for example, has demonstrated the existence of cookeïte in Aalenian black shales. The occurrence of cookeïte allows us to estimate the maximum temperature for the metamorphism at around 325°C. The pressure estimated from fluid inclusion studies (Bernard, 1978; Jenatton, 1981; Nziengui, 1993) is around 200 MPa. This metamorphism is dated at around 26.5 Ma by Nziengui (1993). Thus the NRM carried by extremely stable fine-grained, probably single domain, pyrrhotite (Rochette and Lamarche, 1986 and Rochette et al., 1990) with a Curie temperature between 310°C and 335°C (Fig. 2a), is probably a thermoremanent magnetization (TRM) acquired during post metamorphic cooling. The occurence of antipodal normal and reverse polarities as a function of the altitude especialy in area 1 and 2 (Table 1), supports this interpretation. Moreover, the study of Rochette et al. (1992) in the same black schist facies but in the northern valley of Maurienne has shown the record of successive normal and reverse components versus temperature in the same sample, therefore strongly supporting our contention that NRM is a TRM. Even if a metamorphic CRM (chemical remanent magnetization) was responsible for our NRM, the directional interpretation would be identical.

3. Sampling and measurement

We sampled 310 oriented cores in 44 sites from the Bourg d'Oisans area (Fig. 1), where the Liassic marly limestone facies is very well exposed. For each site we cored between 5 and 14 samples with a portable drill and oriented them with a magnetic compass. Local structural measurements were taken, in order to make corrections for bedding (S0) and cleavage (S1).

NRM measurements were carried out with a Molspin spinner magnetometer initially and after treatment at 150°C. For 1 or 2 specimens per site a progressive thermal demagnetization up to 350°C was carried out with the use of a CTF cryogenic magnetometer. The furnace used was a Pyrox fabrication and allowed us to heat 30 to 40 specimens in the same time. The residual field inside was less than about 10 nT. The susceptibility was measured using MS2 Bartington, after each heating step. Results from 5 sites with very low intensity of magnetization (< 0.05 mA/m), have not been taken into account in this study. Site mean directions for the characteristic remanent magnetization (ChRM) were determined through Fisher (1953) statistics on the direction remaining after demagnetization at 150°C (Table 1). For this calculation, some samples where the magnetization is less than 0.05 mA/m were rejected.

4. The data

The presence of pyrrhotite in the studied samples is indicated by a very high resistance to alternating

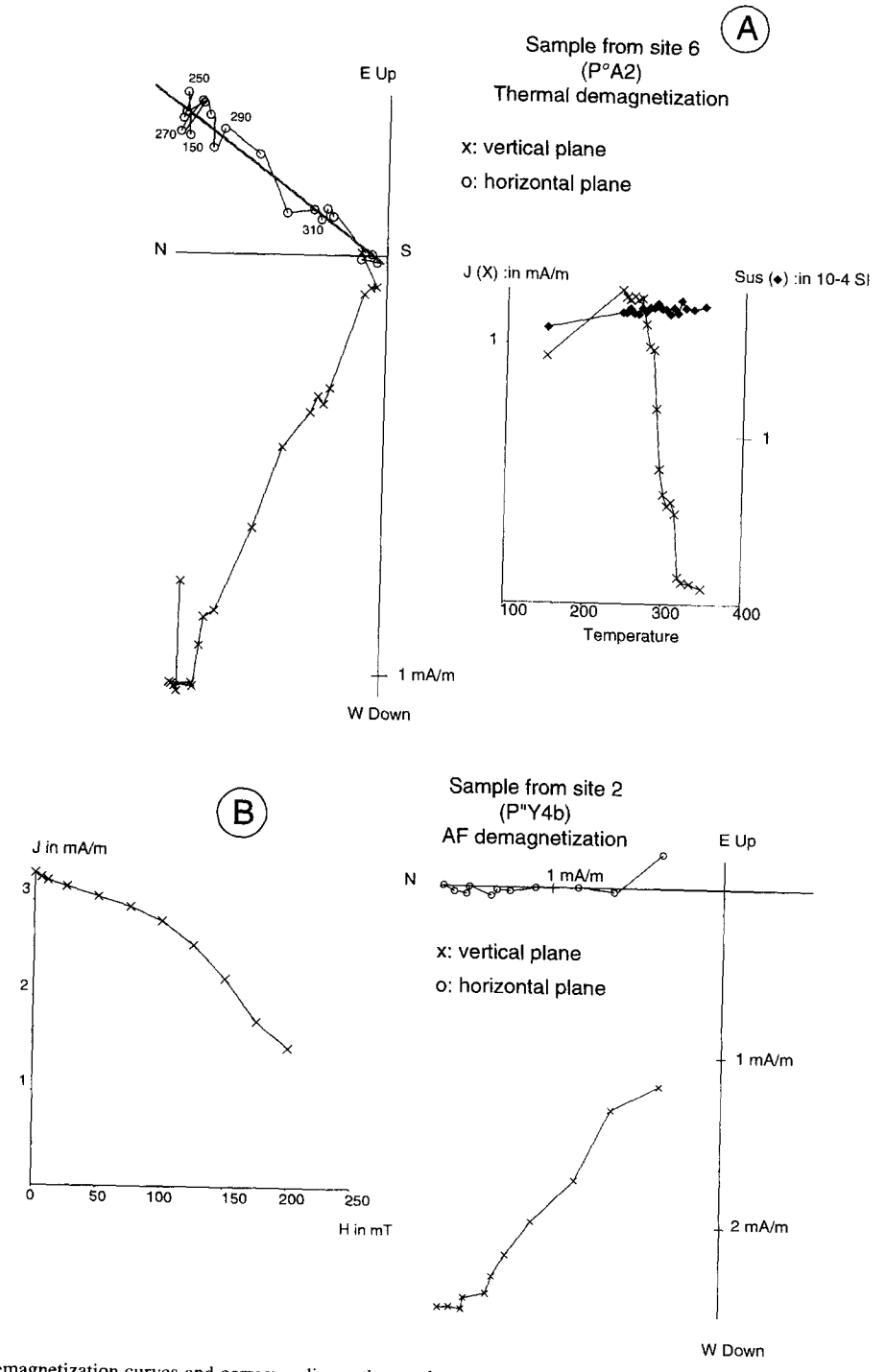


Fig. 2. Stepwise demagnetization curves and corresponding orthogonal projection based on (A) thermal and (B) alternating field treatments In (A) the low field susceptibility after each step is indicated.

field demagnetization and by a Curie temperature of around 325°C (Fig. 2a and b). The absence of changes in the susceptibility during thermal cleaning indicates stability of the magnetic mineralogy up to a temperature of 325°C. The orthogonal plots are linear during the thermal demagnetization between 150°C and the maximum unblocking temperature of 325°C (Fig. 2b). We conclude therefore that possible tiltings or rotations around a vertical axis occured after cooling below 150°C, whereas no significant motion is recorded during cooling from 325° to 150°C.

Table 1
Site mean results. *N*: number of specimens selected for ChRM and *N0* number of specimens studied; *Dec*, *Inc*: respectively geographic declination and inclination of the mean ChRM direction after heating at 150°C; *K* and α95: Fisher's statistical parameters. (1): Site PG from Lamarche et al. (1988)

Site	Alt (m)	Area	*N/N0*	*Dec*	*Inc*	*K*	α95
1	1106	1	8/8	352.5	53.4	245.9	3.5
2	850	1	6/6	359.2	52.9	372.4	3.5
3	900	1	6/6	354.2	54.9	37.6	11.1
4 (1)	780	1	6/?	6.7	53.7	669.0	2.2
5	1010	1	6/6	174.6	−60.5	11.2	20.9
6	1170	1	6/6	21.0	52.9	65.1	8.4
7	1090	2	7/7	36.5	59.4	120.2	5.5
8	1350	2	6/7	38.9	52.4	214.3	4.6
9	1395	2	9/9	236.7	−52.7	39.1	8.3
10	1500	3	8/9	173.6	−60.6	167.1	4.3
11	1500	3	4/6	162.7	−53.0	69.6	11.1
12	1542	3	10/13	167.5	−58.3	24.8	9.9
13	1710	3	6/6	167.2	−51.2	175.7	5.1
14	1510	3	5/5	165.3	−43.7	18.0	18.5
15	1350	3	7/7	168.6	−37.8	11.4	18.6
16	1245	3	12/12	166.6	−52.2	18.1	10.5
17	1120	3	7/7	178.8	−51.6	85.0	6.6
18	1025	3	5/5	169.1	−57.5	78.1	8.7
19	950	3	7/7	191.9	−64.7	8.6	21.8
20	730	4	5/5	193.0	−72.0	18.8	18.1
21	760	4	6/8	6.7	40.4	38.0	11.0
22	820	4	5/7	185.3	−64.0	34.0	13.3
23	910	4	8/8	181.3	−54.8	168.1	4.3
24	950	4	5/5	190.4	−56.7	186.9	5.6
25	1000	4	8/8	179.4	−56.4	66.1	6.9
26	1050	4	6/6	173.7	−55.8	142.7	5.6
27	1100	4	7/7	187.2	−54.3	362.3	3.2
28	1160	4	9/9	196.8	−50.2	58.0	6.8
29	1240	4	6/7	181.9	−54.9	27.8	12.9
30	1300	4	8/8	184.1	−54.0	600.5	2.3
31	1340	4	9/12	162.6	−53.0	46.7	7.6
32	1380	4	6/7	193.6	−55.7	76.1	7.7
33	1530	5	6/7	199.3	−56.1	96.5	6.9
34	1560	5	7/7	194.1	−46.1	96.9	6.2
35	1210	5	6/6	192.0	−58.7	99.8	6.7
36	1290	5	6/6	198.0	−65.6	40.1	10.7
37	1370	5	5/7	172.0	−53.3	155.9	6.1
38	1420	5	7/7	175.5	−61.0	43.7	9.2
39	1510	5	5/6	187.8	−41.4	55.0	10.4
20–32		area 4	88/106	183.6	−55.3	40.7	2.4
33–39		area 5	42/52	188.6	−55.3	37.7	3.6

The ChRM directions became dispersed upon bedding correction. Therefore the fold test is negative. The main penetrative axial plane cleavage is acquired during folding. The magnetization clearly postdates the folds and so the main cleavage (S1 called S2 in Gratier et al., 1973). This penetrative deformation occurs during the late Oligocene. Taking into account the age constraints on metamorphism and tectonic phases, we can propose an age for remanence acquisition. If NRM is a CRM, its age is the one of pyrrhotite formation during the metamorphism which occured at around 26.5 Ma (Nziengui, 1993), that is late Oligocene. If NRM is in fact a TRM, it dates from the post-metamorphic cooling, so we can suggest it began at about the Oligo–Miocene time boundary.

The map of the mean declination (Fig. 1) shows coherent results for each of the five areas. Areas 4 and 5 (Fig. 3a and Table 1), show no significant departure from the expected direction at 20 Ma ($D = 6°$, $I = +58°$) for stable Europe (Besse and Courtillot, 1991). For areas 1, 2 and 3, in contrast, the stereoplot (Fig. 3b), of the NRM mean directions, calculated after thermal cleaning at 150°C, clearly indicates a scatter on both sides of the expected direction. Areas 1 and 3 show an apparent

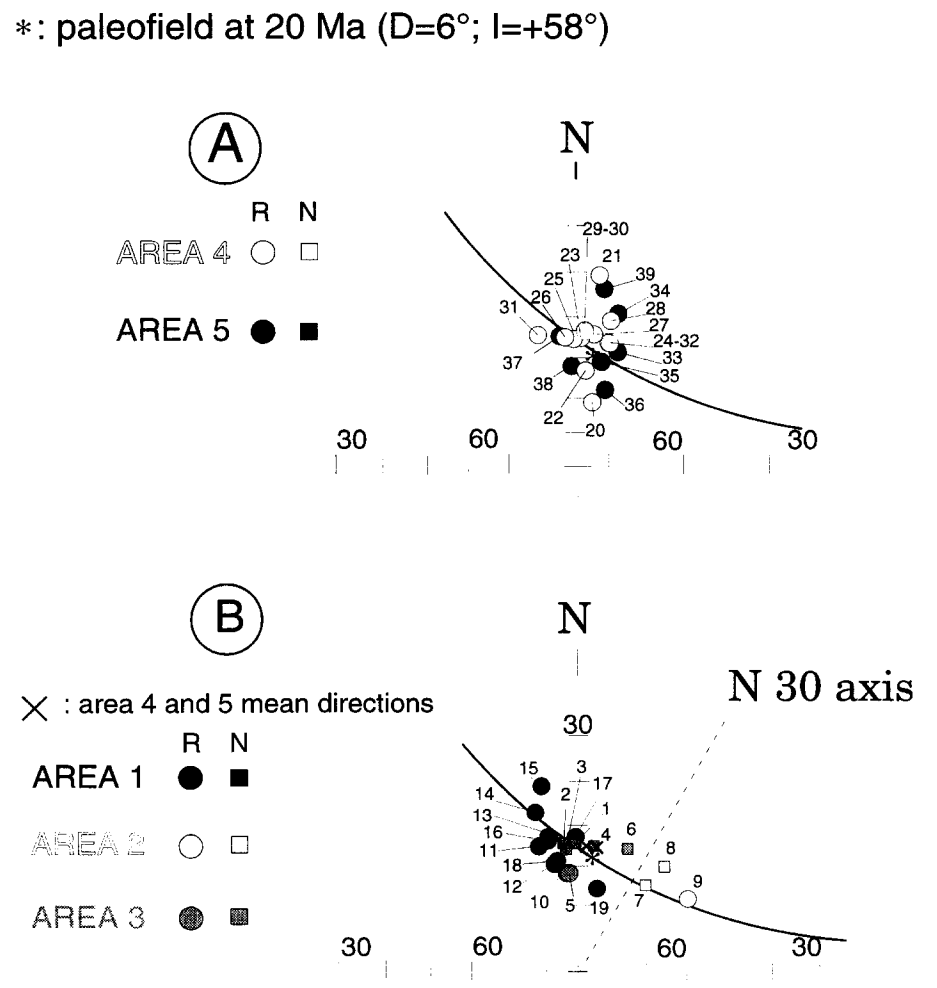


Fig. 3. Equal angle stereoplot (Wulff lower hemisphere) of site mean directions at 150°C (all projected in the same hemisphere). Squares (N): normal polarity, circles (R): reverse polarity.

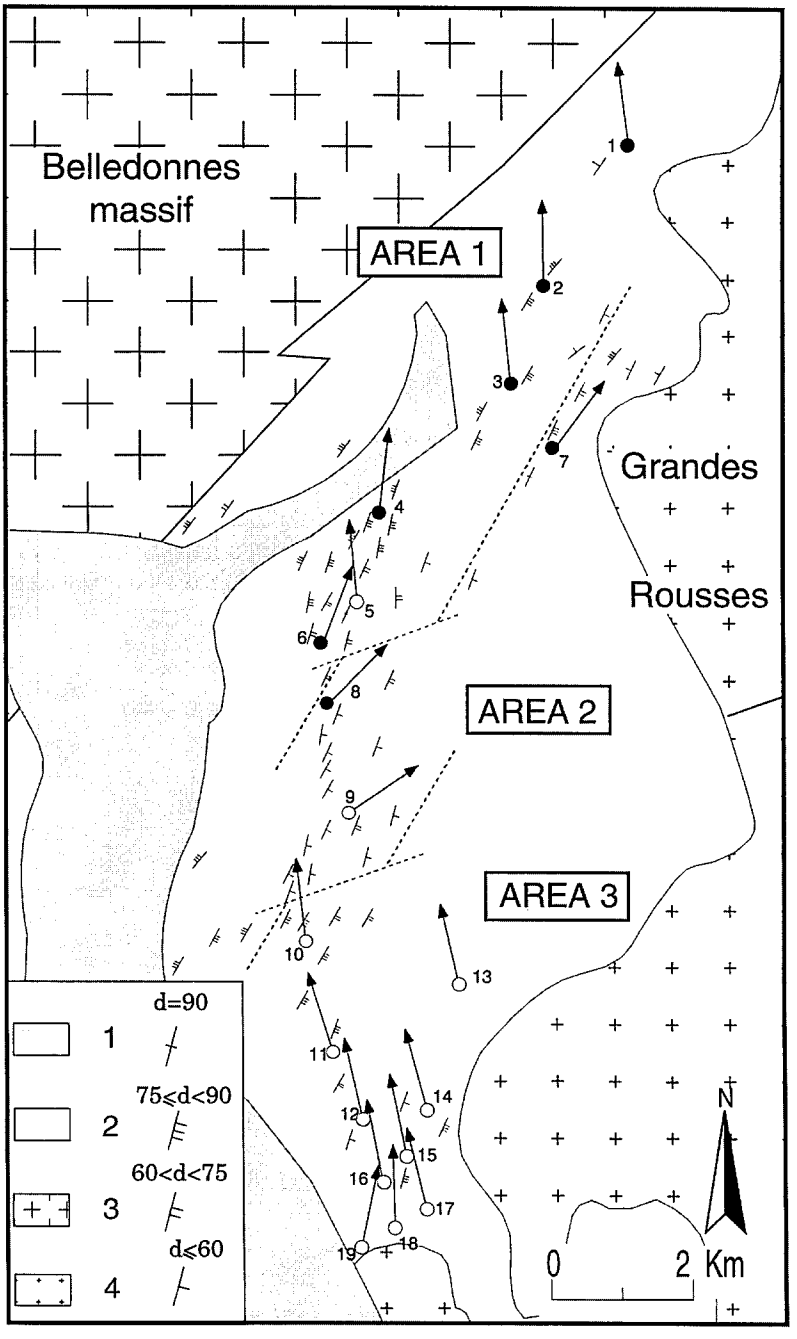


Fig. 4. Comparison between the site mean declinations and main cleavage S1. The regional consistency of the S1 orientation suggests an absence of rotation around a vertical axis. Dashed line: limit of area (1, 2, 3), *d* dip of S1 cleavage, *1* Quaternary, *2* sedimentary cover, *3* Belledonne crystalline basement, *4* Grandes Rousses crystalline basement, circle: reverse polarity, dot: normal polarity.

counterclockwise rotation with respect to the expected direction, while area 2 shows a clockwise rotation.

5. Discussion

The boundaries of these three areas (1, 2, 3) defined, using both declination and S1 measurements (Fig. 4), are oriented approximately SSW–NNE. To be completly in agrement with structural observations we must postulate some local dextral transfert faults not easily visible on the field. Those faults are oriented N60-70E. The same system of faults is well observed in the Grandes Rousses basement (Barfety et al., 1972). The direction of the main cleavage (S1) does not change in the field between areas 1, 2 and 3 (Fig. 4). It is therefore difficult to interpret the motions recorded by paleomagnetism as rotations around only a vertical axis. A combination of rotations around vertical and horizontal axes is always possible, but the dispersion recorded by paleomagnetic data may be also interpreted as due to folding around a horizontal approximatively N20E axis. A detailed microstructural study with bedding and cleavages measurements and relationships was performed in order to verifies that structural planes have been rotated in coherency with paleomagnetic vectors. Unfortunately, we cannot consider S1 cleavage as homogeneous in dip in the whole of the studied area before late folding. This assumption is valid only on a small scale. A fold test taking into account S1 directions is therefore impossible. However when plotting separately for areas 1, 3 and 2 both NRM and S1 pole direction (Fig. 5) it is clear that they have both been folded around the same N20 axis, obtained statistically by a best fitting cone technique. Moreover the average direction show the same tilting of about 25° for structural and paleomagnetic data. The N20E direction is approximatively the main structural direction of the Belledonne crystalline massif. The direction of the Ornon–Col du Sabot fault, which bounds the graben to the northeast and separates it from the Belledonne massif, changes

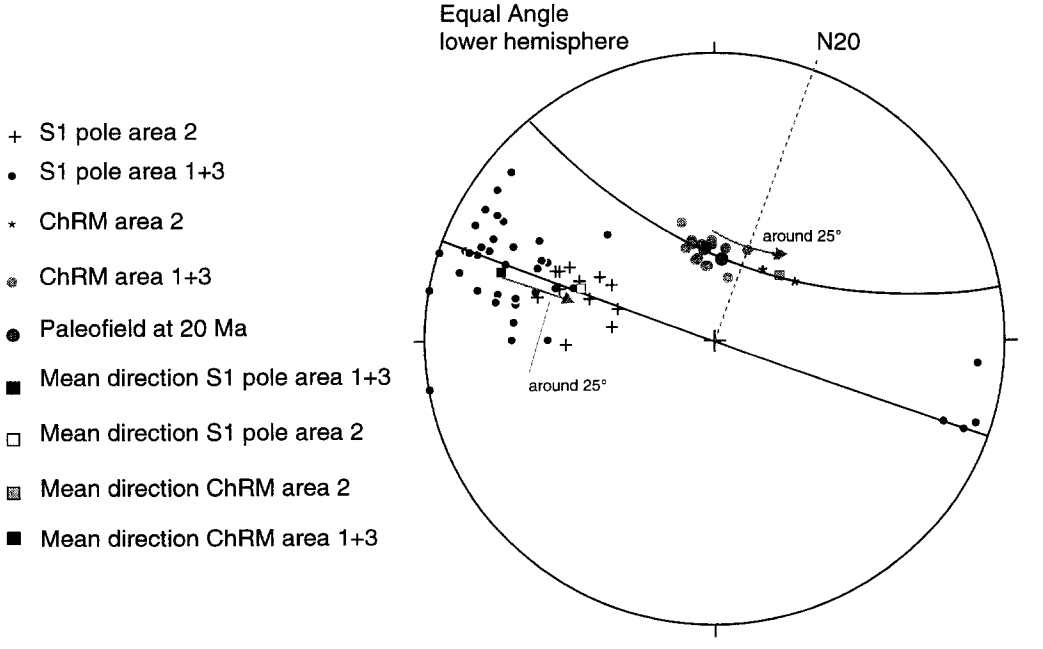


Fig. 5. Equal angle hemispheric projection (Wulff lower hemisphere) of cleaned site mean direction for area 1, 2 (grey dot) and 3 (*), area mean direction (dark grey square: area 1 + 3, pale grey square: area 2), S1 pole direction for area 1, 2 (black dot) and 3 (+) with mean direction (black square: area 1 + 3; open square: area 2). The best small circles fitting with paleomagnetic and structural data are indicated. The tilting around 25° is consistent with all the observations.

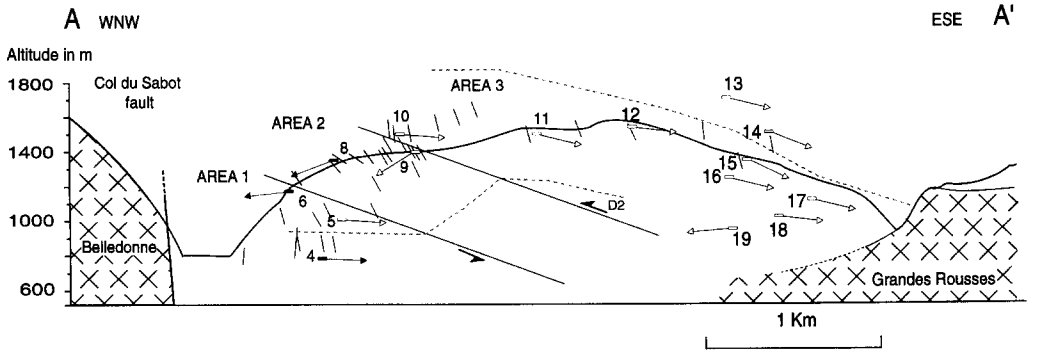


Fig. 6. Schematic cross section AA′ in a N110E plane. Projection of the main cleavage dip (S1) and the tiltings calculated from paleomagnetism shown as angles between the arrows and horizontal; black rectangle and arrow: site of normal polarity, white rectangle and arrow: site of reverse polarity, dashed line: post D1 paleo-horizontal line showing a late fold (= D2).

from N5E in the south to N45E in the northern part of the studied area.

We determined the amount of tilting around the N20E axis for each site through comparison between the observed and the expected directions. The so determined tiltings are projected onto a N110E plane (Fig. 6). Ménard and Rochette (1992) have shown with this method the presence in the northern valley of Maurienne of simple late-stage fold structures. Although these folds have not yet been described in the Bourg d'Oisans area, we suspect their presence. For sites 8 and 9 of area 2 (Fig. 6), the S1 cleavage is less steep than in the adjacent areas 1 and 3 (S0 is not visible). This suggests a tilt of about 20–30° towards the W, which is consistent with the tilts deduced from paleomagnetic and structural data (Fig. 4).

The cleavage data indicate that site 7 (Fig. 7) is located in the overturned western limb of a fold. A comparable tilting of S0 and S1 of about 25° is also evidenced by the paleomagnetic data. Such tilting for sites 7, 8 and 9 is confined and may be attributed to

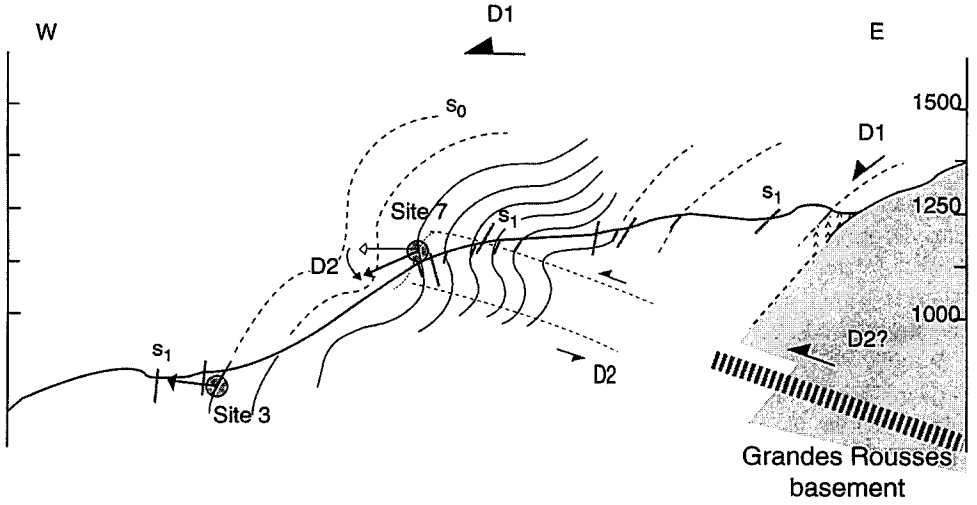


Fig. 7. Cross section BB′ showing consistency between structural and paleomagnetic data and a slight overturning of the fold. Grey dash: projection of S1 dip, black arrow: amount of tilting concluded from paleomagnetism, white arrow: tilting before D2.

a westward shear zone which could be associated with late crystalline thrusting as described by Gratier et al. (1973).

The movements recorded by our paleomagnetic data are younger than about 26.5 Ma; it is possible to specify this age from further observations. Fission tracks dating on apatite in the Belledonne and Grandes Rousses Massifs has given ages between 5 and 9 Ma for cooling below 110°C (Lelarge, 1993; Sabil, 1995). The Zijderveld (1967) plots show the demagnetization of a single component during the thermal demagnetization between 150° and 325°C. Thus we can conclude that no significant motion occurred during cooling from 325° to 150°, and that the tilting must have occurred after cooling below 150°C. Cooling curves deduced from geochronological data in the Mont Blanc area (Soom, 1990) suggest a cooling rate of 10°/Ma. The age of the cooling below the 150° isotherm can thus be estimated at about 9–13 Ma. This is probably the oldest age possible for the tiltings.

Study of syntectonic sediments in the foreland molasse basin dates the beginning of the crustal movement of the Belledonne thrust (Ménard, 1988; Deville et al., 1994) at about the Aquitanian–Burdigalian time boundary (20 Ma). The reality of this thrust, first proposed by Ménard (1979) is now generally accepted from interpretation of the ECORS seismic line (Guellec et al., 1989, 1990; Mugnier et al., 1989; Nicolas et al., 1990). Around 10–11 Ma, deposition of the Upper Freshwater Molasse stopped (Pfiffner, 1986) and this is interpreted as the end of the Molasse basin subsidence. Then we can consider that the External Crystalline Massifs Thrust was active between 20 Ma and at least 10–11 Ma. During all this time the studied area seemed to have been passively carried. According to our age estimated, the late deformation recorded by paleomagnetism is probably related to the end of the External Crystalline Massifs thrust motion. The sedimentary cover folding may be induced by a little basement thrust in the Grandes Rousses massif.

The paleomagnetic data do not clearly argue for counterclockwise rotation as proposed by Gratier et al. (1989) and by Vialon et al. (1989) during the Neogene. A previous paleomagnetic direction obtained by Henry (1992) on the Upper Triassic–Lower Liassic lavas from the same area, shows no significant rotation about a vertical axis. Unfortunately, this direction may be the reflection of superimposed rotations, one during the opening of the Bourg d'Oisans half graben and another during the alpine tectonism. In consequence, it is hazardous to take this direction into account when testing late alpine motions.

6. Conclusions

This study, implementing the previous works of Lamarche et al. (1988) and Ménard and Rochette (1992), indicates the extent to the south of the late folding of the sedimentary cover and dates this movement younger than 9–13 Ma (Upper Miocene). This deformation is probably related to the late history of the thrusting of the Belledonne and Grandes Rousses crystallines massifs. Between 20 Ma and at least 11–10 Ma the Belledonne thrust was active. The Bourg d'Oisans area and the Grandes Rousses massif were passively carried. When the Belledonne thrust stopped, the more internal Grandes Rousses thrust was reactived probably during a short time. This last motion produced a gentle folding in the sedimentary cover probably associated with a westward shear zone. This out-of-sequence senario is consistent with both paleomagnetic and structural observations.

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